Arnoldi Iteration: Krylov Subspaces, GMRES and Ritz Value Approximation

A Non-Symmetric Linear System Solver & Eigenvalue Approximation

Shirsendu Talukder	Biswajit Mondal	Rentu Sk
26MA60R11	26MA91F02	26MA91F05
M.Tech Student	Ph.D. Student	Ph.D. Student

Indian Institute of Technology, Kharagpur

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The Problem & Motivation

Countless scientific and engineering fields demand solving large linear systems $Ax = b$ ($A \in \mathbb{C}^{m \times m}$):

- Optimization & Stability Analysis
- Computational Electromagnetics
- Computational Biology & Physics
- Climate Science
- Quantum Mechanics
- Signal Processing & Machine Learning
- Communication Systems
- Compressive Sensing

Why not direct inversion $x = A^{-1}b$?

- Gauss-Jordan / LU requires $\mathcal{O}(m^3)$ time and heavy memory.
- Destroys sparsity of large matrix A.

Iterative Solution: Krylov Methods

- Construct $x_0, x_1, x_2, \cdots \to x_*$ using only matrix-vector products ($v \mapsto Av$).
- Never forms powers of A or A^{-1}.

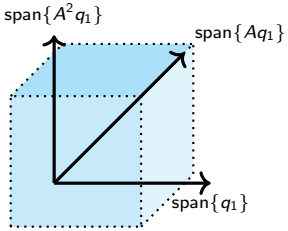
Krylov Subspace: Intuition & Definition

Definition: n^{th} Krylov Subspace

For $A \in \mathbb{C}^{m\times m}$ and $q_1 \in \mathbb{C}^m$, the nth Krylov subspace is:

$$K_n(A, q_1) = \text{span}\{q_1, Aq_1, A^2q_1, \ldots, A^{n-1}q_1\}$$

- **Cayley-Hamilton Intuition:** $A^{-1}b$ lies in the span of $\{b, Ab, \ldots, A^{n-1}b\}$.
- Instead of working with huge $A \in \mathbb{C}^{m\times m}$, we project its action onto an n-dimensional subspace ($n \ll m$).
- Raw vectors $[q_1, Aq_1, \ldots, A^{n-1}q_1]$ become nearly linearly dependent $\implies$ construct an **orthonormal basis** via Arnoldi Iteration.



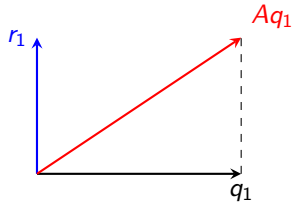
$K_3 = \text{span}\{q_1, Aq_1, A^2q_1\}$

Arnoldi Iteration: Step-by-Step Construction

Arnoldi orthogonalizes $\{q_1, Aq_1, Aq_2, \dots\}$ via Gram–Schmidt starting with $\|q_1\|_2 = 1$.

Step 1:

- Inner product: $h_{11} = \langle q_1, Aq_1 \rangle$ (component along q_1)
- Subtract projection: $r_1 = Aq_1 - h_{11} q_1$
- Residual norm: $h_{21} = \|r_1\|_2 = \|Aq_1 - h_{11} q_1\|_2$
- Normalize (if $h_{21} \neq 0$): $q_2 = r_1 / h_{21} \implies \boxed{Aq_1 = Q_2 \tilde{H}_1}$



Step 2: Process $Aq_2 = h_{12} q_1 + h_{22} q_2 + h_{32} q_3$, where $h_{i2} = \langle q_i, Aq_2 \rangle$:

$$AQ_2 = Q_3 \tilde{H}_2, \qquad \tilde{H}_2 = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \\ 0 & h_{32} \end{bmatrix}_{3\times 2}$$

$$Q_2 = \begin{bmatrix} q_1 & q_2 \end{bmatrix}, \quad \tilde{H}_1 = \begin{bmatrix} h_{11} \\ h_{21} \end{bmatrix}$$

General Arnoldi Relation & Algorithm

After n steps ($Aq_n = \sum_{i=1}^{n+1} h_{in} q_i$):

$$\boxed{AQ_n = Q_{n+1}\tilde{H}_n}$$

$$\tilde{H}_n = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1n} \\ h_{21} & h_{22} & \cdots & h_{2n} \\ & \ddots & \ddots & \vdots \\ & & h_{n,n-1} & h_{nn} \\ & & & h_{n+1,n} \end{bmatrix}_{(n+1)\times n}$$

Why upper Hessenberg? Aq_j uses $q_1, \ldots, q_{j+1}$, introducing one nonzero subdiagonal entry $h_{j+1,j}$.

Algorithm 1: Arnoldi (Modified Gram–Schmidt)

Require: $A \in \mathbb{C}^{m\times m}$, $\|q_1\|_2 = 1$, steps n.
Ensure: $Q_{n+1} = [q_1, \ldots, q_{n+1}]$, $\tilde{H}_n \in \mathbb{C}^{(n+1)\times n}$

1. **For** $j = 1, 2, \ldots, n$:
2. $\quad v \leftarrow Aq_j$
3. $\quad$ **For** $i = 1, \ldots, j$:
4. $\quad\quad h_{ij} \leftarrow q_i^* v; \quad v \leftarrow v - h_{ij} q_i$
5. $\quad h_{j+1,j} \leftarrow \|v\|_2$
6. $\quad$ **If** $h_{j+1,j} = 0$ **then stop**
7. $\quad q_{j+1} \leftarrow v / h_{j+1,j}$

Interpretation: Orthogonal Projection onto K_n

Let $Q_n = [q_1, \ldots, q_n] \in \mathbb{C}^{m \times n}$ have orthonormal columns ($Q_n^* Q_n = I_n$). Without breakdown ($h_{j+1,j} \neq 0$), $K_n = \text{Range}(Q_n) = K_n(A, q_1)$.

Proposition: Orthogonal Projector

$P_n = Q_n Q_n^*$ projects orthogonally onto K_n:

$$P_n^2 = P_n, \quad P_n^* = P_n, \quad (x - P_n x) \perp K_n$$

Since $A(K_n) \not\subseteq K_n$, define $T_n := P_n \circ A|_{K_n}$:

$$\begin{array}{ccc} K_n & \xrightarrow{A|_{K_n}} & \mathbb{C}^m \\ & \searrow^{T_n} & \downarrow P_n \\ & & K_n \end{array}$$

In basis Q_n ($\varphi_n(y) = Q_n y$, $\varphi_n^{-1}(x) = Q_n^* x$), the matrix of T_n is:

$$[T_n]_{Q_n} = \varphi_n^{-1} T_n \varphi_n = Q_n^*(Q_n Q_n^* A|_{K_n}) Q_n = \boxed{Q_n^* A Q_n = H_n}$$

Multiplying $AQ_n = Q_{n+1}\tilde{H}_n$ on the left by Q_n^*:

$$H_n = Q_n^* Q_{n+1} \tilde{H}_n = \begin{bmatrix} I_n & 0 \end{bmatrix} \tilde{H}_n$$

Key Takeaway

$H_n \in \mathbb{C}^{n \times n}$ is $\tilde{H}_n$ with its last row deleted.

Computing Eigenvalues: Ritz Values & Residuals

Eigenvalues θ of H_n are **Ritz values**, and $u = Q_n y$ is the **Ritz vector**:

$$H_n y = \theta y \quad (\|y\|_2 = 1) \implies u = Q_n y \quad (\|u\|_2 = 1)$$

Ritz Residual Formula

From $AQ_n = Q_n H_n + h_{n+1,n} q_{n+1} e_n^*$:

$$\boxed{Au - \theta u = h_{n+1,n}\, q_{n+1}\, e_n^* y}$$

$$\|Au - \theta u\|_2 = |h_{n+1,n}|\, |e_n^* y|$$

Key Properties

- **Breakdown ($h_{n+1,n} = 0$):** K_n is A-invariant; every (θ, u) is an **exact eigenpair**.
- **Galerkin Condition:** $Q_n^*(Au - \theta u) = 0 \iff \boxed{Au - \theta u \perp K_n(A, b)}$
- **Numerical Range:** $\theta = u^* A u \in W(A)$.

Spectral Distance Bound: Normal vs Nonnormal

Proposition: Spectral Bound for Normal Matrices

If A is normal ($A = V\Lambda V^*$, V unitary) and $\|Au - \theta u\|_2 \leq \varepsilon$ with $\|u\|_2 = 1$, then:

$$\boxed{d(\theta, \Lambda(A)) = \min_{\lambda_j \in \Lambda(A)} |\lambda_j - \theta| \leq \|Au - \theta u\|_2 = |h_{n+1,n}|\,|e_n^* y| \leq \varepsilon}$$

Proof: Let $z = V^* u$ ($\|z\|_2 = 1$). Since $Au - \theta u = V(\Lambda - \theta I)z$:

$$\|Au - \theta u\|_2^2 = \|(\Lambda - \theta I)z\|_2^2 = \sum_{j=1}^{m} |\lambda_j - \theta|^2 |z_j|^2 \geq d(\theta, \Lambda(A))^2 \sum_{j=1}^{m} |z_j|^2 = d(\theta, \Lambda(A))^2.$$

Crucial Distinction: Normal vs. Nonnormal

- **Normal A:** Small Ritz residual *guarantees* θ is close to an actual eigenvalue.
- **Nonnormal A:** Ritz values can have tiny residuals while lying **far from the spectrum** $\Lambda(A)$.

Experiment 1: Normal vs. Nonnormal Matrices (Setup)

We compare two 30×30 matrices with starting vector $q_1 = b/\|b\|_2$:

$$A_{\text{normal}} = \text{diag}(1, 2, \ldots, 30)_{30\times 30}$$

$$A_{\text{nonnormal}} = \begin{bmatrix} 0 & 100 & 0 & \cdots & 0 \\ 0 & 0 & 100 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 100 \\ 0 & \cdots & & \cdots & 0 \end{bmatrix}_{30\times 30}$$

$A_{\text{nonnormal}}$ is strictly upper triangular with 100 on the superdiagonal $\implies$ unique eigenvalue $\Lambda(A_{\text{nonnormal}}) = \{0\}$.

Parameter	Normal	Nonnormal
Matrix dimension m	30	30
True Spectrum $\Lambda(A)$	$\{1, \ldots, 30\}$	$\{0\}$
Arnoldi steps n	10	1
Starting vector b	random	$(1, 0.01, 0, \ldots)^T$
Number of Ritz values	10	1

Table 1: Experimental configuration.

Experiment 1: Visual Comparison of Eigenvalues & Ritz Values

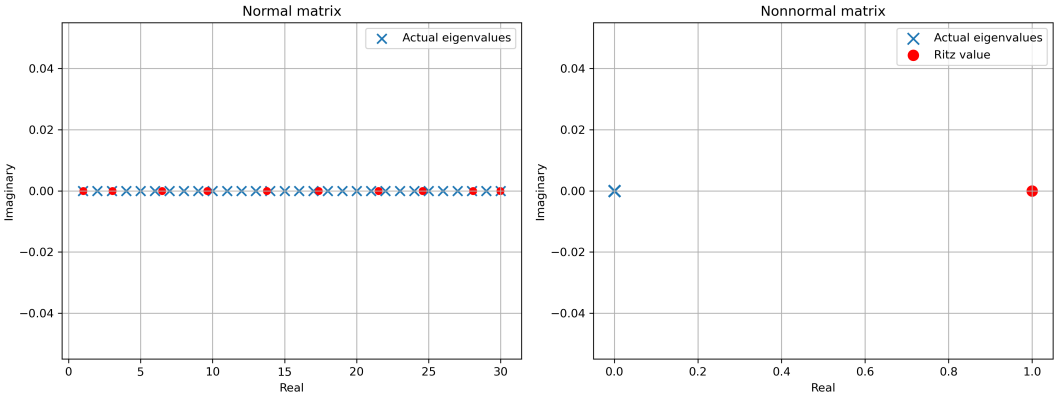


Figure 1: Actual eigenvalues and Ritz values for a normal matrix (left, $n = 10$) and a nonnormal matrix (right, $n = 1$).

Experiment 1: Quantitative Results (Tables 1 & 2)

Table 1: Normal Matrix ($n = 10$ steps)

Ritz value θ	Residual $\|Au - \theta u\|_2$	$d(\theta, \Lambda(A))$
1.006770	1.862×10^{-1}	6.770×10^{-3}
3.048372	8.135×10^{-1}	4.837×10^{-2}
6.485809	2.226×10^{0}	4.858×10^{-1}
9.626454	2.502×10^{0}	3.735×10^{-1}
29.977610	2.936×10^{-1}	2.239×10^{-2}
28.072269	1.062×10^{0}	7.227×10^{-2}
13.768214	3.405×10^{0}	2.318×10^{-1}
17.351938	3.714×10^{0}	3.519×10^{-1}
24.576341	1.994×10^{0}	4.237×10^{-1}
21.516959	2.973×10^{0}	4.830×10^{-1}

Table 2: Nonnormal Matrix ($n = 1$ step)

θ	$\|Au - \theta u\|_2$	$d(\theta, \Lambda(A))$
0.999900	9.999×10^{-3}	9.999×10^{-1}

Inference

- **Normal:** $d(\theta, \Lambda(A)) \leq \|Au - \theta u\|_2$ holds strictly for every Ritz value.
- **Nonnormal:** Residual is 9.999×10^{-3}, but distance to true eigenvalue 0 is 0.9999 ($100\times$ larger!).

GMRES: Least-Squares Formulation

To solve $Ax = b$ with initial guess x_0, residual $r_0 = b - Ax_0$, and $\beta = \|r_0\|_2$, we search in $x_0 + K_n(A, r_0)$:

$$x_n = x_0 + z_n, \quad z_n \in K_n(A, r_0) \implies \boxed{x_n = x_0 + Q_n y} \quad (y \in \mathbb{C}^n,\ q_1 = r_0/\beta)$$

Proposition: GMRES Least-Squares Reduction

$x_n = x_0 + Q_n y_n$ minimizes $\|b - Ax\|_2$ over $x \in x_0 + K_n(A, r_0)$ if and only if y_n solves:

$$\boxed{\min_{y \in \mathbb{C}^n} \|\beta e_1 - \tilde{H}_n y\|_2} \qquad \text{with } \|b - Ax_n\|_2 = \|\beta e_1 - \tilde{H}_n y_n\|_2.$$

Proof: By $AQ_n = Q_{n+1}\tilde{H}_n$, $r_0 = \beta q_1 = Q_{n+1}\beta e_1$, and orthonormal columns $Q_{n+1}^* Q_{n+1} = I_{n+1}$:

$$\|b - Ax_n\|_2 = \|r_0 - AQ_n y\|_2 = \|Q_{n+1}(\beta e_1 - \tilde{H}_n y)\|_2 = \|\beta e_1 - \tilde{H}_n y\|_2.$$

Solved at each step via QR factorization / Givens rotations on the $(n+1) \times n$ Hessenberg matrix $\tilde{H}_n$.

GMRES Algorithm & Breakdown

Algorithm 2: GMRES

Require: $A \in \mathbb{C}^{m\times m}$, $b \in \mathbb{C}^m$, x_0, $n_{\max}$, tol ε.

1. $r_0 \leftarrow b - Ax_0$, $\beta \leftarrow \|r_0\|_2$, $q_1 \leftarrow r_0/\beta$, $n \leftarrow n_{\max}$
2. **For** $j = 1, 2, \ldots, n_{\max}$:
   - $v \leftarrow Aq_j$
   - **For** $i = 1, \ldots, j$: $h_{ij} \leftarrow q_i^* v$; $v \leftarrow v - h_{ij} q_i$
   - $h_{j+1,j} \leftarrow \|v\|_2$
   - $y_j \leftarrow \arg\min_{y\in\mathbb{C}^j} \|\beta e_1 - \tilde{H}_j y\|_2$
   - **If** $\|\beta e_1 - \tilde{H}_j y_j\|_2 \le \varepsilon$ **or** $h_{j+1,j} = 0$: $n \leftarrow j$; **break**
   - $q_{j+1} \leftarrow v/h_{j+1,j}$
3. **Return** $x_n \leftarrow x_0 + Q_n y_n$

Lucky Breakdown ($h_{n+1,n} = 0$)

If $h_{n+1,n} = 0$ and A is nonsingular:

- $AQ_n = Q_n H_n$ and H_n is nonsingular.
- $y_n = \beta H_n^{-1} e_1$ gives $\|r_n\|_2 = 0$, so x_n is the **exact solution**!

Restarted GMRES(m)

As j grows, storage grows as $\mathcal{O}(jm)$ and work as $\mathcal{O}(j^2 m)$. Restarting every m steps caps memory cost.

GMRES as a Polynomial Approximation Problem

Since $z_n \in K_n(A, r_0)$, write $z_n = s_{n-1}(A)r_0$ where $s_{n-1}(t) = c_0 + c_1 t + \cdots + c_{n-1}t^{n-1}$. Then the residual $r_n = b - A(x_0 + z_n)$ is:

$$r_n = \big[I - As_{n-1}(A)\big]r_0 = p_n(A)r_0, \qquad \text{where } \boxed{p_n(t) = 1 - t\,s_{n-1}(t), \quad \deg p_n \le n, \quad p_n(0) = 1}$$

Thus GMRES is equivalent to the polynomial minimization problem:

$$\boxed{\|r_n\|_2 = \min_{\substack{\deg p \le n \\ p(0)=1}} \|p(A)r_0\|_2}$$

Proposition: Convergence Bound for Normal Matrices

If $A = V\Lambda V^*$ is normal (V unitary, $c = V^* r_0$), then $\|p(A)r_0\|_2^2 = \sum_{i=1}^m |p(\lambda_i)|^2 |c_i|^2$, giving:

$$\boxed{\frac{\|r_n\|_2}{\|r_0\|_2} \le \inf_{\substack{\deg p \le n \\ p(0)=1}} \max_{\lambda \in \Lambda(A)} |p(\lambda)|}$$

Spectral Geometry & Nonnormal Convergence Bound

Examples of Spectral Behavior:

1. **Few Distinct Eigenvalues:** If $\Lambda(A) = \{1, 2, 3\}$, $p(t) = (1 - t)(1 - t/2)(1 - t/3)$ vanishes on $\Lambda(A) \implies$ exact solution in ≤ 3 steps.
2. **Clustered Disk Away from 0:** If $|\lambda - \gamma| \leq \rho < |\gamma|$, taking $p_n(z) = (1 - z/\gamma)^n$ yields geometric convergence:

$$\boxed{\frac{\|r_n\|_2}{\|r_0\|_2} \leq \left(\frac{\rho}{|\gamma|}\right)^n}$$

3. **Eigenvalues Near/Around 0:** Since $p_n(0) = 1$, $p_n(\lambda_i) \approx 1$ near 0, retarding convergence.

Proposition: Nonnormal Bound

If $A = X\Lambda X^{-1}$ is diagonalizable:

$$\|p(A)\|_2 \leq \kappa_2(X) \max_{\lambda \in \Lambda(A)} |p(\lambda)|$$

$$\boxed{\frac{\|r_n\|_2}{\|r_0\|_2} \leq \kappa_2(X) \inf_{\substack{\deg p \leq n \\ p(0)=1}} \max_{\lambda \in \Lambda(A)} |p(\lambda)|}$$

where $\kappa_2(X) = \|X\|_2 \|X^{-1}\|_2$.

For highly nonnormal A, $\kappa_2(X) \gg 1$, so eigenvalues alone can be misleading.

Experiment 2: Disk vs. Rose Spectrum (Mathematical Setup)

Let $m = 500$, $G \in \mathbb{R}^{m \times m}$ with i.i.d. $N(0,1)$ entries, $b = (1, \ldots, 1)^T$, and $x_0 = 0$. We run GMRES for $n = 1, \ldots, m$ on two matrices:

1. Disk Spectrum Matrix A

$$A = 2I + \frac{1}{2\sqrt{m}} G$$

- Eigenvalues fill disk $|\lambda - 2| \leq \frac{1}{2}$.
- Center $\gamma = 2$, radius $\rho = \frac{1}{2}$ (0 is outside).
- Candidate rate: $(\rho/\gamma)^n = (0.25)^n$.

2. Rose Spectrum Matrix A_{mod}

$$A_{\text{mod}} = A + D = \text{diag}\left(i \sin(k\theta_j) e^{-i\theta_j}\right) + \frac{1}{2\sqrt{m}} G$$

- $k = 6$, $\theta_j = (j-1)\pi/m$,
  $d_j = (-2 + \sin(k\theta_j)\sin\theta_j) + i \sin(k\theta_j)\cos\theta_j$.
- Eigenvalues lie in a thickened rose curve **surrounding** 0.

Experiment 2: Eigenvalue Spectra (Disk vs. Rose)

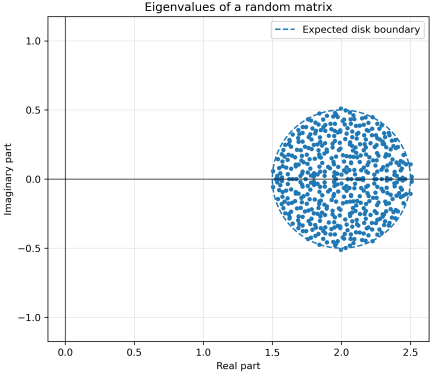


Figure 2: Eigenvalues of A: Clustered in disk $|\lambda - 2| \leq 0.5$ away from 0.

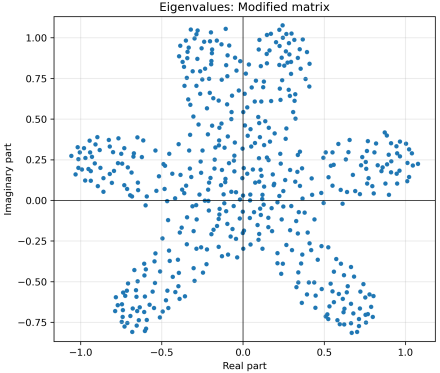


Figure 3: Eigenvalues of A_{mod}: Thickened 6-petal rose surrounding 0.

Experiment 2: GMRES Residual Convergence Comparison

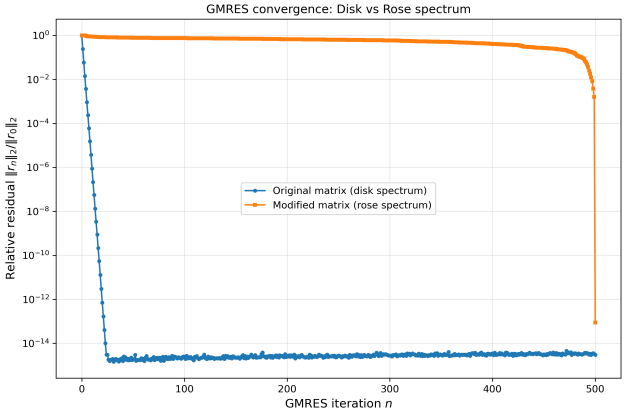
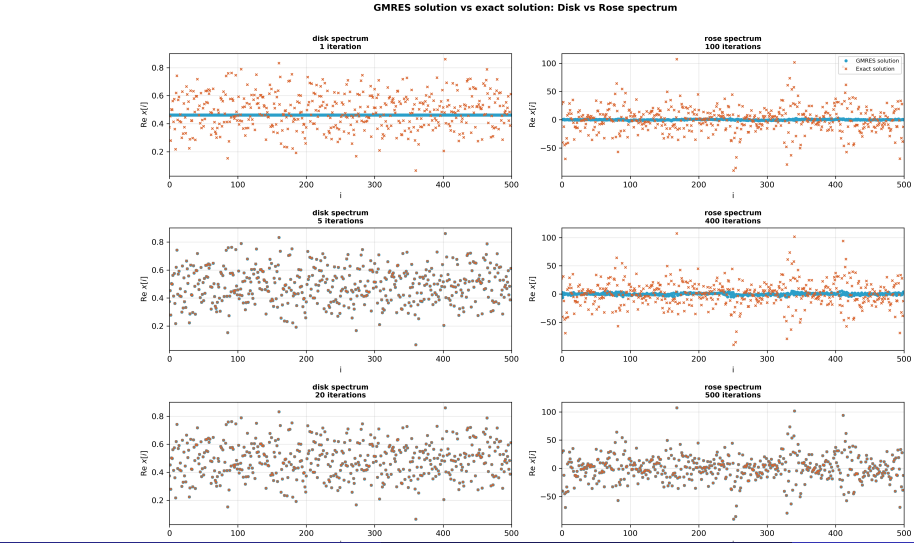


Figure 4: Top: Spectra of A (disk) and A_{mod} (rose). Bottom: Relative residual $\|r_n\|_2/\|r_0\|_2$ vs. 0.25^n.

Key Takeaways

- **Fast Convergence on Disk (A):** Relative residual decays geometrically at the theoretical rate $(\rho/\gamma)^n = 0.25^n$.
- **Conditioning of Eigenvectors:** Due to random G, A is nonnormal with $\kappa_2(X) \approx 5 \times 10^2$, satisfying $\|r_n\|_2/\|r_0\|_2 \leq \kappa_2(X)(0.25)^n$.
- **Slow Convergence on Rose (A_{mod}):** Because the rose spectrum encloses the origin 0 (where $p_n(0) = 1$), no disk bound holds and convergence stalls until n approaches m.

Comparison


GMRES solution vs exact solution: Disk vs Rose spectrum
disk spectrum
1 iteration
rose spectrum
100 iterations
GMRES solution
Exact solution
disk spectrum
5 iterations
rose spectrum
400 iterations
disk spectrum
20 iterations
rose spectrum
500 iterations
Re x[i]
i

Summary & Takeaways

- **Arnoldi Process:** Orthonormalizes the Krylov basis via $AQ_n = Q_{n+1}\tilde{H}_n$ and represents the orthogonal projection of A onto K_n as the upper Hessenberg matrix $H_n = Q_n^* A Q_n$.
- **Ritz Value Approximation:** Eigenpairs of H_n satisfy the Galerkin orthogonality condition $Au - \theta u \perp K_n$ with computable residual $\|Au - \theta u\|_2 = |h_{n+1,n}|\,|e_n^* y|$.
- **Normal vs. Nonnormal Spectra:** Small Ritz residuals guarantee $d(\theta, \Lambda(A)) \leq \varepsilon$ for normal matrices, whereas nonnormal matrices can exhibit small residuals far from the spectrum.
- **GMRES Polynomial Optimality:** Solves $\min_y \|\beta e_1 - \tilde{H}_n y\|_2 \iff \min_{\deg p \leq n, p(0)=1} \|p(A) r_0\|_2$.
- **Spectral Geometry:** Eigenvalues clustered in a disk away from 0 yield geometric convergence $(\rho/|\gamma|)^n$, while spectra surrounding the origin severely slow down GMRES.

Thank You!

Questions & Discussions